HS524M – Urban Informatics and Spatial Analytics

Total Marks: 20

Deadline: 13th October 2026; 17:00

Submission requirement: Both Question 1 and Question 2 must be presented on your personal webpage/portfolio. You must submit (i) the complete code/notebook(s) used for both questions and (ii) the link to the webpage where the outputs and discussion are hosted.

Question 1 – Interactive Web Mapping of Waste-Bin Observations [5 Marks]

Using the waste-bin observations collected through QField, create an interactive Folium web map showing the locations of the recorded waste bins.

Your map should:

- display the waste-bin observations as interactive point features;
- provide a tooltip and/or popup for each observation showing relevant information such as the location/title, description or field note, and other useful attributes;
- display the photograph associated with the observation within the popup;
- clearly identify the student who collected the observation;
- include appropriate map controls, including a Layer Control where relevant;
- be saved as an HTML map and hosted on your personal webpage/portfolio so that it can be accessed interactively through a web browser. Ensure that the photographs are also visible after the map is hosted, not only on your local computer.

You may increase the spatial coverage of your map by incorporating waste-bin observations collected by other students in the class. However, any data obtained from another student must be clearly acknowledged. For such observations, the popup should explicitly state the name of the student who originally collected the data, along with the other observation details and photograph.

The final map should be clear, usable, and allow a viewer to interactively explore the spatial distribution and characteristics of the waste-bin observations.

Submit for Question 1: the code/notebook and the webpage link containing the hosted interactive map.

Question 2 – Individual Google Earth Engine Urban Analysis [15 Marks]

Each student has can choose a unique city–dataset–analysis combination. Complete the assigned analysis using Google Earth Engine (Python/geemap). The tasks are provided in the allocation table in this document and in the shared spreadsheet, please add your name and roll no. to the analysis row you wish to work on. Request to not remove the names of other students.

Assignment spreadsheet: Google Sheet – individual city/task allocation and dataset details

City Boundary

Use an appropriate boundary for your assigned Indian city/Urban Local Body. One suggested source is the Indian urban-boundary release below, which includes Urban Local Body boundaries. You may use another legitimate municipal, state-government, official GIS/open-data, or other reliable boundary source. In all cases, clearly state the boundary source in your submission.

Suggested boundary source: Indian Urban Local Body boundaries (urban release)

What you have to do

1. Download/prepare the boundary of your assigned city and convert it to a suitable GeoDataFrame/Earth Engine object.
2. Load the Google Earth Engine dataset(s) specified for your assignment and apply the required spatial, temporal, and quality filters.
3. Carry out the assigned indicator/change/exposure analysis. For comparisons across years, use a consistent seasonal window and a consistent visualization scale wherever applicable.
4. Produce clear thematic map(s) with appropriate title, legend/colour scale, units and city boundary.
5. Calculate meaningful numerical summary statistics relevant to your question (for example: area change, percentage change, city-wide mean, exposed population, or exposure share).
6. Create at least one dynamic temporal output where the dataset permits it: an animated GIF, time slider, or before–after swipe comparison. The dynamic output must be visible or accessible from your webpage.
7. Discuss the results. Your discussion should explain the main spatial pattern, the temporal change or exposure pattern, and important limitations such as spatial resolution, temporal resolution, cloud/data availability, or differences between datasets.
8. Host the Question 2 outputs and discussion on your personal webpage/portfolio.

Note for population-exposure assignments: If the task uses Sentinel-5P or night-time lights, define 'relatively high' values using a transparent city-specific rule such as the highest quartile unless another rule is specified. Do not describe satellite column measurements as individual or ground-level personal exposure. Report the result as population residing in areas with relatively high satellite-observed values.

Submit for Question 2: the complete code/notebook and the webpage link containing the maps, dynamic output, numerical results, and discussion.

Google Earth Engine datasets used in the assignments

The Earth Engine IDs and key characteristics below are provided to help you locate the correct datasets. Always check the Earth Engine Data Catalog documentation for band meanings, units, scale factors and limitations.

Dataset	What it represents	Earth Engine ID	Resolution / time	Main bands
GHSL Built-up Surface	Built-up surface area per 100 m grid cell; total and predominant non-residential built-up surface.	JRC/GHSL/P2023A/GHS_BUILT_S	100 m; 5-year epochs, 1975–2030	built_surface; built_surface_nres
GHSL Building Volume	Estimated total and non-residential building volume per 100 m grid cell.	JRC/GHSL/P2023A/GHS_BUILT_V	100 m; 5-year epochs, 1975–2030	built_volume_total; built_volume_nres
GHSL Degree of Urbanisation	Settlement classification from rural classes through peri-urban/urban clusters to urban centres.	JRC/GHSL/P2023A/GHS_SMOD_V2-0	1 km; 5-year epochs, 1975–2030	smod_code
GHSL Population	Residential population count per grid cell.	JRC/GHSL/P2023A/GHS_POP	100 m; 5-year epochs, 1975–2030	population_count
VIIRS Night-time Lights	Monthly average night-time radiance; cf_cvg gives cloud-free observation coverage.	NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG	~464 m; monthly, 2014–present	avg_rad; cf_cvg
JRC Global Surface Water	Annual water classification distinguishing not-water, seasonal water and permanent water.	JRC/GSW1_4/YearlyHistory	30 m; annual, 1984–2021	waterClass
Dynamic World	Sentinel-2-derived near-real-time land-cover probabilities/classes for water, trees, grass, flooded vegetation, crops, shrub/scrub, built, bare and snow/ice.	GOOGLE/DYNAMICWORLD/V1	10 m; 2015–present	label; built; trees; crops; bare; water; etc.

Dataset	What it represents	Earth Engine ID	Resolution / time	Main bands
Sentinel-2 Surface Reflectance	Harmonized Level-2A surface reflectance used here to calculate indices other than NDVI.	COPERNICUS/S2_SR_HARMONIZED	10–20 m bands; 2017–present	B2 Blue; B3 Green; B4 Red; B8 NIR; B11 SWIR1; B12 SWIR2
Sentinel-5P NO_2	Satellite-observed atmospheric NO_2 vertical-column information; use tropospheric NO_2 for the assigned exposure tasks.	COPERNICUS/S5P/OFFL/L3_NO2	From 2018; frequent observations	tropospheric_NO2_column_number_density
Sentinel-5P CO	Satellite-observed total carbon-monoxide column.	COPERNICUS/S5P/OFFL/L3_CO	From 2018; frequent observations	CO_column_number_density
Sentinel-5P Aerosol Index	UV absorbing aerosol index; positive values can indicate UV-absorbing aerosols such as dust or smoke.	COPERNICUS/S5P/OFFL/L3_AER_AI	From 2018; frequent observations	absorbing_aerosol_index
WorldPop Population	Estimated residential population in approximately 100 m grid cells.	WorldPop/GP/100m/pop	~100 m; annual, 2000–2021	population
WorldPop Age–Sex	2020 population disaggregated by sex and age group in approximately 100 m grid cells.	WorldPop/GP/100m/pop_age_sex	~100 m; 2020 only	population; M_0 … M_80; F_0 … F_80

Sentinel-2 indices used in selected assignments:

- NDBI = (B11 − B8) / (B11 + B8) → built-up signal
- MNDWI = (B3 − B11) / (B3 + B11) → surface-water signal
- NDMI = (B8 − B11) / (B8 + B11) → vegetation/surface moisture signal
- BSI = ((B11 + B4) − (B8 + B2)) / ((B11 + B4) + (B8 + B2)) → bare-soil / exposed-surface signal

Individual Question 2 allocations

Each row is a unique assignment. Use the city, dataset, period and analysis specified for your assigned row. The shared spreadsheet should be used to mention which row you wish to select.

No.	City	Assigned analysis	Dataset / period	Required output
1	Mumbai	Map 2020 mean tropospheric NO_2. Estimate the number/share of residents aged 65+ living in the city's highest-NO_2 quartile.	S5P NO_2 + WorldPop age–sex; 2020	Monthly 2020 NO_2 GIF/time slider + annual NO_2 map + 65+ exposure map/table + interpretation
2	Kolkata	Map 2020 mean tropospheric NO_2. Estimate exposure of women aged 15–49 to the highest-NO_2 quartile.	S5P NO_2 + WorldPop age–sex; 2020	Monthly 2020 NO_2 GIF/time slider + exposure map/table for women 15–49 + interpretation
3	Chennai	Map 2020 mean CO column. Estimate the number/share of residents aged 65+ in the city's highest-CO quartile.	S5P CO + WorldPop age–sex; 2020	Monthly 2020 CO GIF/time slider + 65+ exposure map/table + interpretation
4	Hyderabad	Map the 2020 absorbing aerosol index. Estimate exposure of children aged 0–14 to the city's highest aerosol-index quartile.	S5P AER_AI + WorldPop age–sex; 2020	Monthly 2020 aerosol GIF/time slider + child-exposure map/table + interpretation
5	Ahmedabad	Map 2020 mean tropospheric NO_2 and compare male versus female population residing in the city's highest-NO_2 quartile.	S5P NO_2 + WorldPop age–sex; 2020	Monthly 2020 NO_2 GIF/time slider + male/female exposure comparison + interpretation
6	Pune	Quantify and map the expansion of total built-up surface.	GHSL Built-up Surface; 1990–2020	5-year built-up time slider/GIF + 1990–2020 change map + total/percentage increase
7	Surat	Quantify how total built-up surface changed over two decades and identify major growth zones.	GHSL Built-up Surface; 2000–2020	5-year time slider/GIF + change map + area/percentage increase
8	Jaipur	Map the spatial expansion of built-up surface relative to the 2005 footprint.	GHSL Built-up Surface; 2005–2020	2005–2020 epoch slider/GIF + expansion map + summary statistics
9	Lucknow	Compare changes in total built-up surface and predominant non-residential built-up surface.	GHSL Built-up Surface; 2000–2020	Time slider/GIF + total vs non-residential change maps/graph + discussion

No.	City	Assigned analysis	Dataset / period	Required output
10	Nagpur	Analyse the spatial and total change in building volume.	GHSL Building Volume; 2000–2020	5-year building-volume GIF/time slider + change map + city totals
11	Indore	Compare total and non-residential building-volume change.	GHSL Building Volume; 1990–2020	5-year slider/GIF + total/non-residential comparison + interpretation
12	Bhopal	Examine how the Degree of Urbanisation classes changed over time.	GHSL SMOD; 2000–2020	5-year classification time slider/GIF + transition/area summary + interpretation
13	Patna	Examine how residential population distribution and total population changed.	GHSL Population; 2000–2020	5-year population GIF/time slider + change map + population totals + discussion
14	Vadodara	Compare the rate of built-up surface growth with population growth.	GHSL Built-up Surface + GHSL Population; 2000–2020	Two time-series summaries + maps + dynamic built-up or population visualization + interpretation
15	Kanpur	Create annual mean night-time-radiance surfaces and analyse the long-term trend.	VIIRS Night-time Lights; 2014–2024	Annual radiance GIF/time slider + city mean-radiance graph + change map + discussion
16	Ludhiana	Create annual mean night-time-radiance surfaces and analyse the long-term trend.	VIIRS Night-time Lights; 2014–2024	Annual radiance GIF/time slider + city mean-radiance graph + change map + discussion
17	Agra	Create annual mean night-time-radiance surfaces and analyse the long-term trend.	VIIRS Night-time Lights; 2014–2024	Annual radiance GIF/time slider + city mean-radiance graph + change map + discussion
18	Nashik	Estimate total population and population residing in the highest quartile of night-time radiance in 2020.	VIIRS Night-time Lights + WorldPop; 2020	Monthly 2020 light GIF/time slider + exposure map + total/exposed population and share + discussion
19	Rajkot	Estimate total population and population residing in the highest quartile of night-time radiance in 2020.	VIIRS Night-time Lights + WorldPop; 2020	Monthly 2020 light GIF/time slider + exposure map + total/exposed population and share + discussion
20	Varanasi	Create annual mean night-time-radiance surfaces and analyse the long-term trend.	VIIRS Night-time Lights; 2014–2024	Annual radiance GIF/time slider + city mean-radiance graph + change map + discussion
21	Faridabad	Calculate NDBI and examine how the built-up spectral signal changed. Use the same Jan–Mar seasonal window each year.	Sentinel-2 SR; 2018–2025	Annual Jan–Mar NDBI GIF/time slider + 2018–2025 difference map + city mean NDBI trend

No.	City	Assigned analysis	Dataset / period	Required output
22	Meerut	Calculate MNDWI and examine changes in the surface-water signal. Use the same Oct–Dec seasonal window each year.	Sentinel-2 SR; 2018–2025	Annual Oct–Dec MNDWI GIF/time slider + 2018–2025 difference map + interpretation
23	Ranchi	Calculate NDMI and examine changes in the moisture signal. Use the same Mar–May seasonal window each year.	Sentinel-2 SR; 2018–2025	Annual Mar–May NDMI GIF/time slider + change map + annual mean trend
24	Coimbatore	Calculate BSI and examine changes in bare/exposed-surface signal. Use the same Jan–Mar seasonal window each year.	Sentinel-2 SR; 2018–2025	Annual Jan–Mar BSI GIF/time slider + 2018–2025 difference map + interpretation
25	Kochi	Compare NDBI and MNDWI between two years to discuss built-up and surface-water signals. Use the same Jan–Mar window.	Sentinel-2 SR; 2018 vs 2025	Before–after swipe or GIF + NDBI and MNDWI change maps + numerical summaries
26	Visakhapatnam	Quantify the annual area of permanent surface water and examine long-term change.	JRC Global Surface Water; 2001–2021	Annual water GIF/time slider + permanent-water area graph + change map
27	Vijayawada	Quantify the annual area of seasonal surface water and examine long-term variability.	JRC Global Surface Water; 2001–2021	Annual GIF/time slider + seasonal-water area graph + interpretation
28	Srinagar	Compare permanent and seasonal surface-water area through time.	JRC Global Surface Water; 1990–2021	Annual water GIF/time slider + two-class area graph + discussion
29	Guwahati	Identify areas transitioning among non-water, seasonal water and permanent water.	JRC Global Surface Water; 2000–2021	Annual time slider/GIF + transition map + class-area summary
30	Thiruvananthapuram	Compare change in built land and tree cover.	Dynamic World; 2018–2025	Annual time slider/GIF + built vs tree change maps/graph + discussion
31	Bhubaneswar	Compare change in built land and tree cover.	Dynamic World; 2018–2025	Annual time slider/GIF + built vs tree change maps/graph + discussion
32	Raipur	Compare change in built land and tree cover.	Dynamic World; 2018–2025	Annual time slider/GIF + built vs tree change maps/graph + discussion
33	Mysuru	Compare change in built land and tree cover.	Dynamic World; 2018–2025	Annual time slider/GIF + built vs tree change maps/graph + discussion

General presentation and interpretation requirements

- All maps should be legible and use meaningful titles, legends/colour scales and units.
- For temporal comparisons, keep the colour scale and seasonal window consistent across years unless there is a justified reason not to.
- Do not interpret a visual change without also reporting numerical evidence.
- Discuss data limitations. In particular, acknowledge differences in spatial resolution when combining an environmental raster with WorldPop.
- Cite the city-boundary source and the main dataset source(s) used.
- Your webpage should present the analysis as a short, coherent story: question → method → map/dynamic output → numerical result (if any) → interpretation.

Earth Engine Data Catalog: https://developers.google.com/earth-engine/datasets/catalog/